Pranav Dubey

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Lucknow, Uttar Pradesh

Education

Vellore Institute of Technology, Bhopal 2023 – 2027 (Expected)
B.Tech in Computer Science Engineering CGPA: 8.07/10

Technical Skills

Programming	Python
GenAI & LLMs	LLM Integration, RAG, Prompt Engineering, LangChain, Ollama
Machine Learning	PyTorch, TensorFlow, Keras, Scikit-learn, CNNs, LSTMs, YOLOv8
Backend & APIs	FastAPI, Streamlit
Data & Databases	NumPy, Pandas, Matplotlib, SQL
DevOps & Tools	AWS, Docker, Git, Postman

Projects

Congenital Heart Defect Prediction System Live Demo | GitHub
Tech Stack: Python, PyTorch, FastAPI, PostgreSQL, Docker

- Trained a dual-branch ConvNeXt to detect ASD and VSD in chest X-rays (82.6% accuracy, 0.83 macro F1).
- Applied CLAHE contrast enhancement, feature fusion and focal loss in the training pipeline.
- Developed a FastAPI backend with 22+ REST APIs on PostgreSQL, containerized via Docker.
- Launched a live web demo serving model predictions and patient history records end-to-end.

AI Video Surveillance: Anomaly Detection System Live Demo | GitHub
Tech Stack: Python, PyTorch, YOLOv8, OpenCV, FastAPI, Docker

- Engineered a ResNet50 + LSTM anomaly detector on 1.37M+ frames from 1,900 UCF-Crime videos.
- Achieved 0.803 AUC-ROC, beating the Sultani et al. (2018) baseline of 0.754, with Grad-CAM explanations.
- Mitigated a 76/24 class imbalance with ranking loss and weighted sampling; added YOLOv8 object context.
- Deployed via an async FastAPI job API on Hugging Face Spaces, backed by 48 automated tests.

AI Email Agent Live Demo | GitHub
Tech Stack: OpenAI API, Whisper, Gmail API

- Created a GenAI email agent that turns typed or spoken requests into Gmail emails via OAuth 2.0.
- Raised recipient accuracy from 19.2% (regex) to 94.2% with GPT-4o-mini at $0.06 per 1,000 emails.
- Cut LLM-invented content from 83.5% to 0% and Whisper hallucinations from 93.3% to 3.3%.
- Flagged 100% of uncertain recipient addresses with zero false alarms on a 64-input benchmark.

Newscheck: Cross-Source News Verification GitHub
Tech Stack: Python, LangChain, ChromaDB, Ollama, FastAPI

- Architected a local RAG system that separates independent news coverage from syndicated wire copy.
- Designed a syndication detector with F1 0.999 and 0 false merges across 19,959 real article pairs.
- Lifted trimmed wire-copy recall from 0% to 100% with a containment signal (overall 77.9% to 99.8%).
- Reached 81.7% grounded stance accuracy with Mistral-7B; shipped with 228 tests at 79% coverage.

Certifications

- **NPTEL Marketing Analytics**, IIT Kharagpur: Scored 97%, top 5% of certified learners.
- **Google IT Support Certificate**, Google (2026)
- **The Bits and Bytes of Computer Networking**, Coursera (2025)